

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2023

Course: B. Tech. Branch : Electronics and Computer Engineering/Electronics and Computer Science Engineering

Subject Code & Name: BTESC305 Digital Electronics and Microprocessor **Semester : III**

Max Marks: 60

Date: 21/08/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve any two of the following.		12
A) Describe De Morgan's theorem and its application in simplifying Boolean expressions.	L6	6
B) Solve following questions based on number systems. i. Represent $(-42)_{10}$ using 8-bit signed binary notation (2's complement). ii. Convert the decimal number 125 to binary. iii. Convert the hexadecimal number A7F to binary.	L3	6
C) Perform following logical operations between 11110101 and 10101010. i. OR ii. NAND iii. Exclusive – OR	L3	6
Q.2 Solve any two of the following.		12
A) Simplify the Boolean expression $G(A, B, C) = \Sigma(0, 1, 2, 4, 5, 6)$ using a 3-variable K-Map.	L6	6
B) Implement the boolean expression $F(A, B, C) = \Sigma m(0, 1, 3, 5, 7)$ using a multiplexer.	L4	6
C) Explain how a parity checker/generator is used to detect and correct errors in data transmission.	L5	6
Q. 3 Solve any two of the following.		12
A) Explain the conversion of J-K Flip-Flop into D Flip-Flop.	L6	6
B) Describe the operation of a parallel to serial converter. Where is it commonly used?	L6	6
C) Discuss the various applications of counters in digital systems.	L5	6

Q.4 Solve any two of the following.		12
A) Draw pin diagram of an 8085 microprocessor. Briefly explain any four pins of it.	L4	6
B) Draw timing diagram of following instructions. i. MVI A, 55H ii. STA 526AH	L1	6
C) With suitable examples of each explain any three addressing modes of 8085 microprocessor.	L3	6
Q. 5 Solve any two of the following.		12
A) With neat diagram explain architecture of 8086 microprocessor.	L1	6
B) Find physical address for following values of CS & IP registers of 8086 microprocessor. i) CS: 1220H IP: 0243H ii) CS: 1F00H IP: 0003H iii) CS: AA00H IP: 1212H	L5	6
C) What is DMA controller? Briefly explain 3 modes of data transfer in DMA.	L3	6

*** End ***